Step 1: Interaction Mechanics & Input Paradigm Mapping

- **User Flow Selection**:
    - Flow selection: An accessible multi-page checkout process on Zara.com.
    - “See Shopping Cart” page: This is the first step in the multi-page checkout process and it shows the items added to the cart, the total amount due before taxes, and a “Continue” button.
    - Sign-In/Guest Checkout options: The second page shows a login page that gives you the options to sign into your current account, create a new account, or continue as “Guest”.
    - Delivery Method Page: The 3rd step is choosing where you would like the merchandise delivered. The 3 options are an address of a residential place, any local store locations, or a “Delivery Point” of your choice (this is a business address, found this out after clicking it).
    - Order Summary page: The 4th step takes us to the Order Summary page where it gives you a preview of everything you ordered and your delivery method selection.
    - Selecting Payment Method page: This next step is selecting how you would like to pay for your apparel. It gives 8 different options including credit card, paypal, gift card, apple, google pay, etc. Then after selecting a payment method, payment detail input boxes appear below to enter your payment method information. Then you click finalize payment to take you to the next page.
    - Finalize Purchase Page: This last step shows you an overview of all the options you selected and the merchandise you wish to purchase along with your payment information and delivery method. Then you would click “Finalize Payment” in order to create your order.
- **Input Mechanics Analysis**:
    - Touch Gestures: This would primarily exist on mobile/tablet mediums. This includes vertical scrolling through your order information, tap on selections/buttons you would like to push, and horizontal swiping on carousels of products within your cart.
    - Pointing Devices: This would primarily exist on desktop since the pointing devices reference a mouse input. Hovering over items

states thumbnail previews/navigation links and mouse clicking for CTA buttons.
    - Keyboard Focus Paths: This is a tab order for form inputs that ensure that screen reader users and users that only use a keyboard as input can complete the purchase without feeling like they are trapping their focus in any way.
    - Screen Reader/Voice Inputs: This is where ARIA labeling for forms, fieldsets, and alerts (errors/warnings) should exist so that assistive technology can assist impaired users.
- **Fitts' Law Target Sizing**:
    - Primary Action Buttons: The primary action buttons include “Continue” and “Finalize Payment”. They measure at a height of 40px but we need them to exceed 48px in order to fully satisfy Fitt’s Law.
    - Form Control & Inputs: These form attributes render automatically according to screen size because they are responsive. This is good but not 100% aligned with Fitt’s Law. We need the min-height and min-width to be 48px in order to satisfy Fitt’s Law.
    - Spacing & Target Isolation: There is adequate spacing between secondary buttons and primary conversion buttons. This helps prevent mis-taps.

Step 2: High-Fidelity Prototyping & Micro-Interaction Design

- **Interactive Prototype Creation**:
    - https://www.figma.com/design/UDqqgjq2DqU3VnprKAULge/Untitled?node-id=34-1641&t=CkUuRKJTJbQAFLCm-1
    -
- **State Transition Definitions:**

Component	**State Transition**	**Trigger Event**	**Animation / Timing Spec**	**Programmatic / Accessibility Hook**
CHECKOUT	Default → Resting	None	None	aria-label="checkout"
CHECKOUT	Default	Hover	Transform from #FFFFF to #00000	aria-label="checkout

BTN	→ Hover		with #FFFFF border	"
Ship To Me Box	Default → Focus	TAB press/Mouse Click	#FFFFF Border applied	aria-label="ship-to-me"
Select Payment Method	Default → Active	TAB press/Mouse Click	#FFFFF Border applied	aria-label="payment-method"
Finalize Payment Btn	Disabled	None	Credit Card information needs to be filled before button is active	aria-label="finalize-payment"
Finalize Payment Btn	Active → Loading	Enter press/Mouse Click	Pop up that states "Payment Finalizing"	aria-label="finalize-payment"
Expiration Input box	Default → Error	Inputting invalid input	#D0342C Warning MM/YY accepted only	aria-label="expiration-box"

Step 3: Accessibility Architecture & WCAG Specification

- **Semantic HTML Markup**:
    - Screen 1: Click CHECKOUT
        - <section aria-label=”checkout”>
        - <button class=”blk-btn”>CHECKOUT</button>
    - Screen 2: Your Bag
        - <section aria-label=”ship-to-me”>
        - <div class=”active d-flex”>Ship To Me
 <input type=”radio” name=”shipOption”><label for=”shipOtpion”>Sunday $4.99</label>
 <input type=”radio” name=”shipOption”><label for=”shipOtpion”>Tomorrow $19.95</label>
    - Screen 3: Sign In
        - <section aria-label=”sign-in”>
        - <form action="action_page.php" method="post">Sign In</form>...
    - Screen 4: Official Checkout page
        - <section aria-label=”checkout-overview”>
        - <h1>CHECKOUT</h1> Contact Info ...

        - Screen 5: Payment page
            - <section aria-label=”payment-method”>
            - <h1>Select Payment Method</h1> ...
        - Screen 6: Loading Payment Processing page
            - <section aria-label=”finalize-payment”>
            - <h1>Payment Processing</h1> ...
        - Screen 7:
            - <section aria-label=”payment-confirmation”>
            - <h1>THANK YOU FOR YOUR ORDER!</h1> ...
- ARIA & Screen Reader Hooks:
    - Specify explicit `aria-*` attributes and live region notifications for dynamic updates:
        - Use `aria-expanded="true/false"` for collapsible UI elements.
            - Collapsible UI elements are each payment method with different input for different selections
        - Use `aria-live="polite"` or `aria-live="assertive"` for dynamic status messages and inline form validation errors.
            - aria-live=assertive would be used in the error of input boxes such as entering an invalid month and year for the payment information.
        - Ensure logical keyboard tab order (`tabindex="0"`) across all interactive widgets.
            - Keyboard users can navigate through the checkout process by using tab, enter, and arrow keys
- Visual Accessibility:
    - The text colors on my prototype match the minimum contrast ratio of 4 5:1 because it has classic high-contrast black elements.
    - The UI and boundary contrast a minimum 3:! Ratio against a white background so that the boundaries are clearly visible.
    - The visible focus for keyboard navigation is accessible because the user can access buttons, links, and form inputs by using the tab and tab + shift combinations.

Post-Mortem Reflection (Metacognition)

Include written responses to the following two reflection questions in the final section of your specification report:

1. How did explicitly defining every component state (Hover, Focus, Active, Loading, Error) during the specification phase prevent dynamic usability bugs that static wireframes completely fail to address?

   This prevents dynamic usability bugs because it demands you to go through each component state and this creates a testing phase to check each one of these off one by one.

2. Describe a scenario in your prototype where system latency or asynchronous data processing required specialized micro-interaction feedback (e.g., loading skeletons or `aria-live` announcements). How does your design keep the user informed and prevent repeated submission errors?

   A scenario where my prototype has asynchronous data processing is when the finalize payment button is clicked and a finalize payment pop up appears to let the user know what is going on. This tells the user that we are in the loading and processing state. Then, once it is completed, a “Thank you for your order” page appears to show completion.